Intellect Design Arena
LMS v22 Study Guide

eMACH.ai Lending Ecosystem
(LOS · LMS · CLMS · Collections · Integration)

Prepared for Intellect Design Arena Training
Based on public Intellect documentation + standard LMS architecture
August 2026

Important Notice

- LMS = Loan Management System (post-disbursement servicing), NOT a learning/training LMS.
- Version 22 is your company's internal release label. Server URLs, credentials, and v22-only screen differences must come from your trainer or internal wiki.
- Do NOT share internal server IPs, passwords, or client bank data outside Intellect.
- Add your own training screenshots under Section 11 (Screenshot Checklist).

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1. What is LMS v22?

Term	Meaning at Intellect
LMS	Loan Management System — post-disbursement servicing
v22	Internal product release (e.g. 22.x). Marketing name: eMACH.ai Retail/Commercial Loan Management
Legacy name	Earlier deployments: Polaris Intellect LMS
Business unit	iGCB (Intellect Global Consumer Banking) for retail

LMS owns: booking → disbursement → interest → dues → repayment → delinquency → accounting → closure.

LMS does NOT own: loan application and credit committee approval (that is LOS).

2. Platform Architecture

eMACH.ai Design Pillars:

Pillar	What it means for LMS
Microservices	Booking, interest, EOD, delinquency as separate deployable services
API-first	1200+ open APIs; LMS integrates via APIs not hard-coded links
Cloud-native	On-prem, private cloud, public cloud (AWS/Azure/GCP), hybrid
Headless	UI (DEP/CBX-R) separate from LMS logic
Event-driven	Loan booked, payment received, DPD crossed → events to other systems
Embedded AI	PF Credit Digital Experts on Purple Fabric

Architecture Diagram (Text):

```
CUSTOMER (Mobile / Web / Branch)
        |
        ▼
   DEP / CBX-R  (Digital journeys: B21–B2C)
        |
        ▼
   LOS (G22) ──► CLMS (G23, G24)
        |
        ▼ (Approved loan handoff)
   LMS v22 (G25 Booking, G26 Servicing, G28 EOD)
        |
        ▼
   iTurmeric / Olive Fabric (APIs, Events, Flows)
```

```
        |
        ▼
Core Banking · GL · Bureau · KYC · CRM · DWH
```

DIY / Customization Stack:

Tool	Purpose
Product Engine	Clone products, masters, parameters
Business Rules Engine (DMN)	Policies, scorecards, delinquency rules
Business Central (BPMN 2.0)	Credit workflows, approvals
Credit Customization Studio	Screens, domain objects, events, alerts
iTurmeric Studio	New APIs, orchestration, Fluid Pages (UI)
Reporting	Standard reports, dashboards, extracts

3. All Related Modules

Stage	Module	Product	Screen Codes
10	Application initiation	LOS	G22
20	Credit decisioning	LOS + Rules + AI	—
30	Approval & monitoring	LOS + CLMS	G23, G24
50	Product management	Product Engine	G21
Booking & servicing	LMS v22	Loan Management	G25, G26
70	Collateral & limits	CLMS	G23, G24
80	Debt / collections	Debt Management	G27 + Collections
Digital	DEP / CBX-R	Digital Engagement	B21–B2C
Batch	LMS batch	Batch Processes	G28
Reports	Reporting	Analytics	K211–K21C
Interfaces	Integration	iTurmeric	K221–K226

3.1 LOS — Loan Origination System

Purpose: Apply → verify → score → approve → prepare for booking.

- Omni-channel sourcing (self, RM, broker, API)
- Data capture: customer, loan, collateral, PD notes
- Verification, assessment, policy, scorecard, delegation matrix
- Workflow: parallel / sequential / STP, SLA/TAT
- Handoff to LMS: sanitized, validated loan data on approval

3.2 LMS v22 — Loan Management System

```
LMS v22
├── Product Creation (setup/masters)
├── Transactions (day-to-day ops)
├── EOD / Batch (automated daily)
```

```
├── Booking/Initiation (G25)
├── Maintenance
└── Queries & Reports
```

3.3 CLMS — Collateral & Limit Management

Collateral: property, vehicle, equipment, inventory, securities, receivables, guarantees.

Lifecycle: earmark, freeze/defreeze, pooling, swap, revaluation, release.

Limits: hierarchical (group → entity → product → facility), sanctioned vs drawing limits.

3.4 Debt Management / Collections

Flow: Data upload → segmentation → worklist → collector assignment → settlement / legal / repossession.

3.5 DEP / CBX-R (Digital Front-end)

Code	Screen
B21	Portfolio summary
B22	Account details
B23	Account statement
B24	Loan repayment
B25	Recent transactions
B26	Interest certificate
B27	Disbursement schedule
B28	New loan request
B29	Disbursement request
B2A	Reschedule request
B2B	Top-up request
B2C	Closure simulation

4. End-to-End Credit Lifecycle Flow

4.1 Happy Path — New Retail Loan

- Customer: B28 New Loan Request via DEP/CBX-R
- LOS: KYC, bureau, score, policy checks
- CLMS: Limits & collateral validation
- LOS: Credit approval
- LMS G25: Retail Loan Booking (handoff from LOS)
- LMS: Disbursal (single/multi-tranche), repayment schedule generation
- Monthly: B24 Loan Repayment → Scheduled Payment → priority allocation → GL
- Nightly G28 EOD: accrual, DPD stamping, GL postings

4.2 Stage Numbering

Stage	Name	Primary System

10	Application initiation	LOS
20	Credit decisioning	LOS
30	Approval & monitoring	LOS + CLMS
40	Advance credit analysis	LOS
50	Product management	Product Engine
60	Field operations	Mobile/LOS
70	Collateral & limits	CLMS
80	Debt management	Collections

4.3 LOS → LMS Handoff (Critical)

- Sanitized payload sent on approval (customer, product, amount, tenor, rate, collateral, schedule)
- LMS performs Retail/Corporate Loan Booking (G25)
- No duplicate data entry in operations
- Failure point: approved in LOS but not booked in LMS — ops must re-trigger or manual booking

5. LMS Internal Structure

5.1 Product Creation

Component	What you configure
Product Definition	Loan category (mortgage, PL, auto, WC)
Rate Type Scheme	Fixed, floating, tier, hybrid, adjustable
Due knock-off priority	Tax → fees → penal → interest → principal
Fee Scheme	Processing, prepayment, late fees
Credit Scheme	NPA/classification rules
Penal Scheme	Penalty on overdue
Delinquency Scheme	DPD buckets, movements
Accounting Scheme	GL mapping, accrual rules

5.2 Booking / Initiation (G25)

Function	Description
Entity Definition	Legal entity / branch structure
Project Definition	Project finance linkage
Retail Loan Booking	Book approved retail loan
Corporate Loan Booking	Book SME/corporate facility
Top-up loan	Additional on existing loan
Rollover loan	Renew maturing facility
Disbursal	Release funds (single/multi beneficiary)
Document creation	Loan agreement, advice
Notepad	Ops notes on account
Payout	Fund transfer to customer/vendor

5.3 Transactions (G26)

Transaction	When used
Scheduled Payment	EMI / installment receipt
Payment Reversal	Reverse wrong posting
Excess Adjustment	Apply surplus to principal or refund
Adhoc Fees & Waivers	One-off charge or waiver
Manual Delinquency	Force/correct DPD bucket
Closure (Maturity & Early)	Normal or prepayment closure
Loan & Disbursal Rescheduling	Change tenor/EMI/disbursement plan
Deferral	Skip/postpone installment (moratorium)
Waivers	Interest/fee waiver

5.4 Maintenance

Function	Purpose
Associate Details	Co-borrower/guarantor updates
Collateral Details	Link/update security
Disbursal Hold	Block further disbursement
Interest Rate Change	Manual or benchmark reset
Link-Delink Fees	Attach/detach fee types
POS Change	Principal outstanding adjustment
Repayment Mode	NACH, standing instruction, cash
Holiday Master	Non-working days for due dates
Bulk Uploads	Mass payments, deferrals

5.5 EOD / Batch (G28)

Batch Job	What it does
Interest calculation	Compute interest for the day/period
Accrual – Due creation	Create installment dues
Late payment fees	Apply penal charges on overdue
Delinquency movement	Move accounts across DPD buckets
Auto Closure	Close zero-balance/matured loans
Auto Reschedule	Apply predefined reschedule rules
Provision calculation	Regulatory provisioning (NPA)
Financial postings	Push entries to GL/sub-ledger

5.6 Delinquency (G27)

Concept	Meaning
DPD	Days Past Due — days after due date unpaid
MPD	Months Past Due (some products)
Accrual stopping	Stop recognizing income on NPA
Income reversal	Reverse accrued interest on classification
Provisioning	Set aside amount per regulator rules

6. Screen & Function Codes

6.1 G-codes — Back-office Lending

Code	Module
G21	Products / Product Engine
G22	Origination (LOS)
G23	Customer Limits
G24	Collateral Management
G25	Loan Initiation / Booking
G26	Loan Servicing (core LMS transactions)
G27	Delinquency Management
G28	Batch Processes (EOD)

6.2 B-codes — Digital Engagement (DEP)

Code	Screen/Function
B21	Portfolio summary
B22	Account details
B23	Account statement
B24	Loan repayment
B25	Recent transactions
B26	Interest certificate
B27	Disbursement schedule
B28	New loan request
B29	Disbursement request
B2A	Reschedule request
B2B	Top-up request
B2C	Closure simulation
B1	Multi-factor authentication
B2	User profiling
B3	Multi-level entitlements
B4	Personalisation
B5	Workflow
B6	Alerts & notifications

6.3 K-codes — Reports & Interfaces

Code	Function
K211	Standard reports
K212	Accounting entries query
K213	Pre-closure simulation
K214	Compliance reporting
K215	Central bank reporting
K216	Repayment reports
K217	Approval & disbursal reports
K218	Collateral reports
K219	Operational analytics

K21A	Accrual reports
K21B	Breach reports (collateral & limits)
K21C	Delinquency reports
K221	Accounting entries → GL
K222	Credit score interface
K223	Rates / FX interface
K224	Multi-channel payments
K225	Customer / accounts data
K226	KYC & blacklist

7. Product Configuration

Policy layers attached to each product:

Policy	Controls
Interest policy	Rate, margin, reset frequency
Fee policy	Processing, maintenance fees
Priority policy	Payment allocation order
Delinquency policy	Bucket rules, NPA triggers
Accounting policy	GL accounts, accrual method
Penalty charges policy	Late payment penalties

Personal Loan Setup Order:

- 1. Create base product (Retail → Unsecured → Fixed → EMI)
- 2. Define rate scheme (e.g. 12% fixed)
- 3. Define fee scheme (e.g. 2% processing)
- 4. Define priority scheme (Fees → Interest → Principal)
- 5. Define delinquency scheme (30/60/90 DPD buckets)
- 6. Define accounting scheme (GL codes)
- 7. Create specific product (min/max amount, tenor, prepayment rules)
- 8. Test: simulator → booking → disbursal → payment → EOD

8. Daily Transactions vs EOD Batch

```
DAY (Online / Branch / API)
  • Disbursal, Scheduled Payment, Reschedule, Waiver
  • Immediate validation + partial accounting

NIGHT (G28 Batch / EOD)
  • Interest accrual for all active loans
  • Due creation for due dates
  • Delinquency stamping (DPD update)
  • Late fees, provisions, consolidated GL postings
```

```
Payment Allocation Example (Priority Scheme):
  Payment ₹10,000 received
    → ₹500  Late fee
    → ₹200  Other charges
    → ₹3,000 Interest due
    → ₹6,300 Principal
```

9. Integration Layer

Integration	Typical Flow
LOS → LMS booking	API event on approval
LMS → Core GL	K221 accounting entries
LMS → Payments	K224 multi-channel payments
Bureau	K222 during origination
KYC	K226 blacklist check
DEP → LMS	B24 repayment API
Olive Fabric	Custom orchestration flows

```
External: Tax DB · ROC · Credit Bureau · E-Sign · Bank Statement Analyzer
                        ↓
                iTurmeric (API + Events)
                        ↓
          LOS · LMS · CLMS · Core · CRM · DWH · ECM · SMS/Email
```

10. Server & Environment Access

Ask your trainer for this access checklist:

Item	What to request
LMS UI URL	Training/SIT environment URL
LOS URL	Same or separate portal with role-based menu
Weaver/iTurmeric	Integration studio URL
Environment	DEV / SIT / UAT / TRAINING
Credentials	Trainee ID + role (Ops, Product Admin, Developer)
VPN	Required for internal IPs (e.g. 172.16.x.x)
Database	Read-only access (Oracle/PostgreSQL)
API gateway	Swagger/OpenAPI docs for LMS v22 APIs
Batch monitor	G28 EOD job status screen
Log location	Application logs for failed booking/payment

Typical Roles:

Role	Can access
Product Admin	G21 product setup

Loan Ops	G25 booking, G26 transactions
Supervisor	Waivers, reversals, manual DPD
Batch Admin	G28 EOD monitoring
Developer	iTurmeric, API config
Read-only	Queries, reports (K21x)

11. Study Checklist & Screenshots

Capture these screenshots during training:

#	Screenshot Topic	Why
1	LMS main menu (G21–G28)	Module map
2	Product Definition screen	Product engine
3	Rate / Fee / Delinquency schemes	Policy setup
4	Retail Loan Booking (G25)	LOS handoff
5	Disbursal screen	Fund release
6	Repayment schedule after booking	EMI structure
7	Scheduled Payment (G26)	Day transaction
8	Payment allocation breakdown	Priority scheme
9	Reschedule / Deferral screen	Modifications
10	Delinquency bucket view (G27)	DPD
11	EOD batch monitor (G28)	Batch jobs
12	GL/accounting entries (K212)	Finance link
13	CLMS collateral link (G24)	Security
14	Limit utilization (G23)	Exposure
15	DEP B24 repayment journey	Channel integration

Topics to Master:

- ☐ Difference between LOS and LMS
- ☐ What happens in EOD vs online transaction
- ☐ Payment priority allocation order
- ☐ Fixed vs floating rate reset
- ☐ Single vs multi-tranche disbursal
- ☐ DPD buckets and NPA classification
- ☐ Top-up vs rollover vs reschedule
- ☐ CLMS collateral lifecycle (earmark → release)

- ☐ LOS → LMS handoff data flow
- ☐ G25 vs G26 vs G28

12. Glossary

Term	Definition
Booking	Create live loan account in LMS after approval
Disbursal	Actual fund transfer to customer
Accrual	Recognize interest income/expense over time
Due	Installment amount owed on a date
DPD	Days Past Due
NPA	Non-Performing Asset
STP	Straight-Through Processing (no manual step)
Moratorium / Deferral	Payment holiday
POS	Principal Outstanding
GL	General Ledger
CAM	Credit Appraisal Memo (corporate)
Drawing power	Amount customer can draw against limit/collateral
PF Credit	Purple Fabric AI experts for lending

13. Official References (Public)

- Retail Loan Management: https://www.intellectdesign.com/solutions/lending-solutions/retail-loan-management/
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